Generative AI for NLP

Comprehensive Study Notes

1. What Generative AI Does for NLP

- Enables machines to **comprehend and generate** human-like language
- Incorporates **context awareness** for coherent interactions
- Goes beyond words — senses **feelings and intentions**
- Uses **predictive analytics** and advanced modeling

2. Evolution of Generative AI for NLP

Stage	Description	Limitation
Rule-Based Systems	Followed predefined grammar/linguistic rules	Rigid, no flexibility
Machine Learning	Statistical methods, learned from datasets	Limited with complex nuances
Deep Learning	Neural networks trained on extensive data	More nuanced interpretation
Transformers	Designed for sequential data, deep context understanding	Current state-of-the-art

3. Key Applications

Machine Translation: More precise, context-aware language conversion between languages

Chatbots / Virtual Assistants: Natural, empathetic, personalized conversations with users

Sentiment Analysis: Grasps subtle language expressions for deeper insights into user sentiments

Text Summarization: Captures core meaning and significance for precise summaries

4. Large Language Models (LLMs)

What they are:

- Foundation models using AI + deep learning on massive datasets (websites, books)
- Called "large" due to training data size — can reach **petabytes**
- Contain **billions of parameters** (variables defining model behavior)
- Fine-tuned during training to optimize performance on specific tasks

What they can do:

- Understand language structure and context comprehensively
- Predict next words in a sequence

- Generate creative content with minimal task-specific training
- Pre-trained generically → fine-tuned for specific industries

5. Key LLM Examples

Model	Architecture	Best For
GPT	Decoder only	Text generation, chatbots
BERT	Encoder only	Sentiment analysis, Q&A
BART	Encoder-Decoder	Versatile NLP tasks
T5	Encoder-Decoder	Versatile NLP tasks

6. GPT vs ChatGPT

	GPT	ChatGPT
Focus	Diverse text generation	Conversation generation
Training	Supervised learning	Supervised + Reinforcement learning
Human Feedback	Not incorporated	Uses RLHF

7. Important Caveats

- LLMs can generate text that **sounds accurate but isn't** (hallucination)
- Must address **biases** in training data
- Consider **societal impact** of generated content

Key Takeaway: Most LLMs are transformer-based. They can be pre-trained generically and fine-tuned for specific use cases — making them highly versatile across industries.